

# Attention Mechanisms for Neural Machine Translation

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## Abstract

We propose a novel attention mechanism that improves translation quality. Our model achieves a BLEU score of 34.2 on the WMT2014 dataset, outperforming prior work by 2.1 points. The approach is simple and efficient [12].

## 1. Introduction

Neural machine translation has advanced rapidly in recent years. The transformer architecture (Vaswani et al., 2017) removed recurrence entirely. Given an input sequence, the model computes attention weights over all positions. This allows the network to capture long-range dependencies effectively and to parallelize computation across the sequence length during training.

Figure 1: The attention weight matrix visualized as a heatmap.

## 2. Method

We define the attention score between query and key vectors. The softmax normalizes these scores into a probability distribution. Each output is a weighted sum of value vectors. We train with the Adam optimizer and a learning rate schedule that warms up over the first 4000 steps and then decays.

## 3. Results

Our method outperforms all baselines on both datasets. Table 1 reports BLEU and chrF scores. Ablations confirm each component contributes.